

Performance Testing

Maximum Marks: 3 Marks

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 ACOE	Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

ML Model Performance Testing

All four classification algorithms were evaluated on an 80/20 stratified train-test split of the 5,000-row flood dataset. Results are reported below.

S.No	Algorithm	Train Accuracy	Test Accuracy	Precision	Recall	F1-Score	Deployed
1	Decision Tree	82.40%	78.17%	0.78	0.78	0.78	No
2	K-Nearest Neighbours	85.50%	80.67%	0.81	0.81	0.80	No
3	Random Forest	88.00%	82.00%	0.82	0.82	0.82	No
4	XGBoost	99.20%	96.55%	0.97	0.97	0.96	Yes

Web Application Testing — Test Cases

All functional test cases were executed manually. Results are documented below.

TC ID	Test Case	Input	Expected Output	Result
TC-01	High-risk flood scenario	annual_rainfall=3500, humidity=92, river_level=8.5	Flood Chance page (red)	PASS
TC-02	Low-risk safe scenario	annual_rainfall=800, humidity=52, river_level=2.1	No Flood page (green)	PASS

TC ID	Test Case	Input	Expected Output	Result
TC-03	Empty form submission	All fields blank	Validation error displayed	PASS
TC-04	Negative input value	river_level = -5	Validation error displayed	PASS
TC-05	Prediction history logging	3 predictions submitted	3 rows in history table	PASS
TC-06	About page architecture	Navigate to /about	Architecture diagram shown	PASS
TC-07	Mobile responsiveness	Open on 375px mobile viewport	Responsive layout correct	PASS